MILITARY SPECIFICATION SHEET

CABLE, SPECIAL PURPOSE, ELECTRICAL, TWO INDIVIDUALLY SHIELDED PAIRS OF 22 (7 X 30) AWG

NOTE: NOT FOR AEROSPACE USE.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-49285.

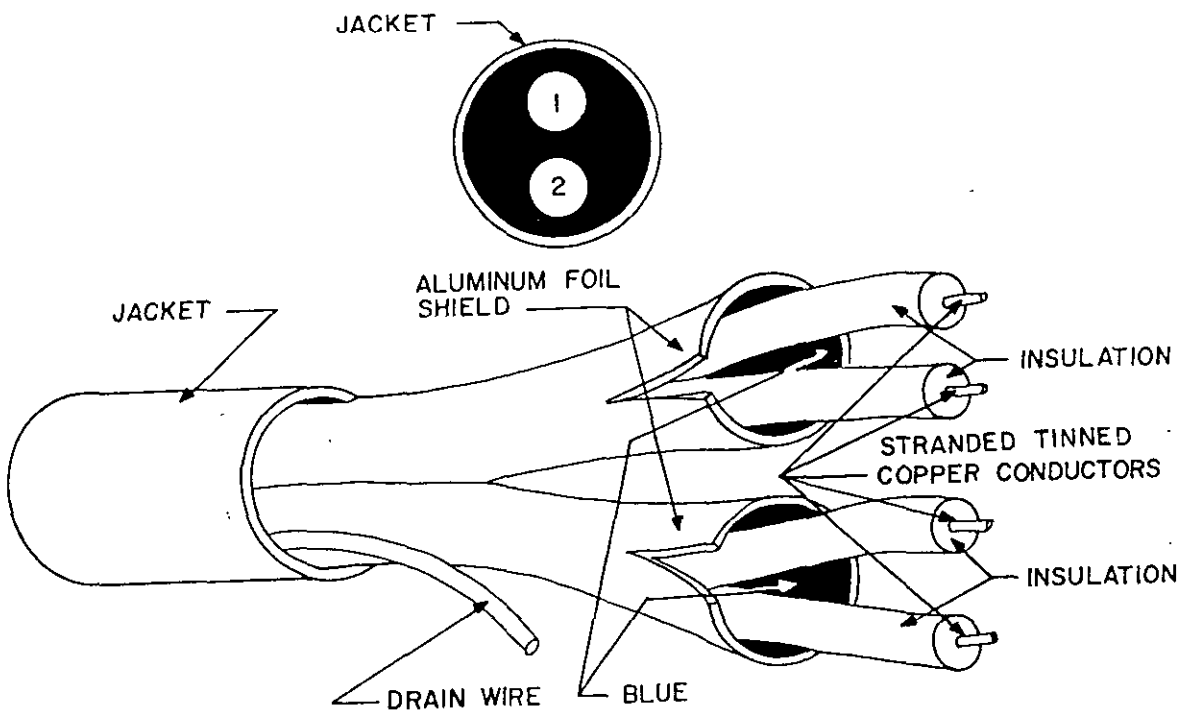


FIGURE 1. Cable illustration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Engineering information: (See table I).

TABLE I. Description, electrical.

Electrical characteristics	Requirements
Nominal capacitance between conductors	35 pF/ft (picofarads/foot)
Nominal capacitance between the conductor and the shield with one conductor tied to the shield	62 pF/ft
Continuous working voltage	400 V rms maximum between conductors
Shield dc resistance	16.7 ohms/1,000 feet maximum

TABLE II. Description, physical.

Components	Construction details
Number of pairs	Two individually shielded
Conductor type and wire size	Stranded tinned copper 22 AWG
Conductor stranding	7 X 30 AWG (16 X 34 AWG optional)
Conductor insulation	Polypropylene
Conductor insulation thickness	0.008 inch nominal thickness
Drain wire type and size	Stranded tinned copper 24 AWG
Drain wire stranding	7 X 32 AWG
Jacket material	Polyvinyl chloride
Jacket thickness	0.019 inch
Finished cable diameter	0.174 inch maximum outside diameter
Cable style (UL)	Not applicable
Tensile strength (jacket)	2,000 pounds per square inch minimum
Elongation (jacket)	150 percent minimum
Overall cabling lay length	8.0 twists per foot ±10 percent

Cabled pair orientation: The position of the red and green pair shield shall determine the rotation direction for pair location (see figure 1).

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REQUIREMENTS:

Design and construction: (See table II).

Shield color code. The shield color shall be blue.

Shield location and orientation. The aluminum is to be located on the outer circumference of the shield, with the polyester side inward. Insulation on the interior of the shield is to be complete with no aluminum available for contact.

Drain wire location. Each drain wire is to be spirally located between the shields and jacket. It is to be in continuous contact with the aluminum surface of the foil shield throughout the cable. The drain wire is stranded (7 X 32) 24 AWG tinned copper. A single drain wire is to be common to both pairs. One drain wire is used in this construction.

Cable temperature rating. The cable temperature rating shall be -20°C to +80°C.

Shield integrity test.*

Frequency	Response **
100 kHz	113 dB below reference
500 kHz	102 dB below reference
1 MHz	96 dB below reference
5 MHz	82 dB below reference
10 MHz	74 dB below reference
15 MHz	70 dB below reference
20 MHz	68 dB below reference
25 MHz	65 dB below reference
30 MHz	63 dB below reference

* Any two readings may deviate to a value of 0.9 times the stated limits.

** All values are a minimum numerical value.

Fixture resonance will occur between 40 and 70 MHz.

At 100 MHz: At least 46 dB below reference.

NOTE: Insert entire cable sample into test fixture. Jacket removal not required.

Flammability: The cable shall pass UL 1581 Vertical-Tray flame test requirements.

Crosstalk test limits. Crosstalk testing is not required for this configuration.

Cabling. The two pairs shall be cabled on a common axis.

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Conductor color code.

a. Pair number one

1st conductor	Black
2nd conductor	Red

b. Pair number two

1st conductor	Green
2nd conductor	White

Durometer hardness. The cable jacket shall have a "Shore A" hardness of 82 ±5.

Part or Identifying Number (PIN): The PIN shall be M49285/13.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Army - MI
Air Force - 71
DLA - ES, IS

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 6145-1114-12)

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